

Why Urges Pass

The Science Behind the QUM Protocol — A Plain-English Guide

This is a friendlier version of a scientific literature review on the QUM (Quick Urge Management) protocol — a three-step method for getting through cravings and behavioral urges. The original review is academic and dense. This version explains the same science with everyday examples, clear language, and all the original citations preserved so you can dig deeper if you want to. The science covered comes from five areas: how cravings actually work in the brain, why mental imagery feeds them, how social media exploits the same circuits, what breathing and exercise really do to your nervous system, and how repeated practice rewires the response over time.

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1. The Big Picture — What QUM Is and Why It Might Work

QUM stands for **Quick Urge Management**. It's a three-phase response you run when a craving or urge hits — whether that's the pull to scroll, smoke, drink, gamble, eat junk, or open the porn tab. The idea is that you don't have to *defeat* the urge with willpower. You just have to *outlast* it, intelligently, while it burns itself out.

The three phases are:

- **Phase 1 — Calm the body.** A short burst of intense movement, then slow controlled breathing. This shifts your nervous system out of "panic" mode.

- **Phase 2 — Hijack the mind.** A demanding visual-spatial task (like sketching a weird shape, doing a puzzle, or playing Tetris). This steals the mental "screen space" your brain needs to keep the craving vivid.
- **Phase 3 — Stay steady, build resilience.** Keep breathing calmly while the wave passes. Each time you survive a wave without giving in, your brain quietly updates its expectations.



Quick analogy: the wave

A craving works like a wave. It rises, peaks, and falls — usually within 10–20 minutes if you don't feed it. Most people try to *fight* the wave (which is exhausting and often fails) or *suppress* it (which makes it bounce back harder). QUM is more like a surfboard — you ride it out while doing things that quietly shrink it.

This guide walks through the science behind why each phase works. The research is real and published in peer-reviewed journals. Some of it is rock-solid; some is still being debated. We'll be honest about which is which.

2. Wanting vs. Liking — Why Cravings Aren't Really About Pleasure

The discovery that changed addiction science

For decades, people assumed cravings were about chasing pleasure — you want the thing because the thing feels good. Then in the early 1990s, two neuroscientists named Kent Berridge and Terry Robinson made a discovery that quietly broke that assumption (Robinson & Berridge, 1993).

They found that the brain has **two separate systems** for reward:

- **"Wanting"** — runs on dopamine, in deep parts of the brain. This is the *pull*, the urge, the need to *get* the thing.
- **"Liking"** — runs on different chemicals (mainly opioids), in tiny specific spots. This is the actual *pleasure* when you have the thing.

Here's the wild part: **these two systems can come apart.** In addiction, the "wanting" system gets hypersensitive — louder and louder over years — while the "liking" system actually goes numb. People still *crave* intensely, but the thing they crave gives them less and less pleasure (Berridge & Robinson, 2016).



The fast food example

Think about the last time you ate junk food you didn't really enjoy. You may have *wanted* the chips badly, eaten them quickly, and then thought, "Wait — that wasn't even that good. Why did

"I want them so much?" That's the wanting-liking gap in action. Your dopamine system pushed you toward them; your pleasure system never delivered the payoff your urge promised.

The proof: rats who wanted but couldn't enjoy

In animal studies, Robinson and Berridge (1993) wiped out almost all dopamine in rats. The rats stopped *going for* rewards entirely — they wouldn't even feed themselves. But if you put sweet liquid directly in their mouths, they still showed normal "this tastes good" reactions. They lost the *wanting* but kept the *liking*.

And the reverse happens in addiction. In a now-famous human study, Fischman and Foltin (1992) gave cocaine users tiny doses of cocaine that were too small to feel anything pleasurable. Users said neither button (cocaine vs. saline placebo) did anything for them — yet they kept pressing the cocaine button anyway. Wanting without liking. The pull was there even when the pleasure wasn't.

A 10-year longitudinal study of people who developed alcohol use disorder found the same pattern: their *wanting* of alcohol steadily climbed over the decade, while their *liking* of it stayed roughly flat (Robinson & Berridge, 2024). The urge grew. The pleasure didn't.

Why this matters for QUM



The key reframe

A craving is not a signal that you would actually enjoy the thing. It's not a real "I want to be happy" message from your true self. It's a **mechanical dopamine spike** — a sensitized neural circuit firing because it learned to fire. Berridge and Robinson (2016) put it bluntly: the brain circuit for wanting is "dissociable from circuitry that mediates the degree to which it is 'liked'."

This reframe matters because shame and self-blame are huge triggers for relapse. If you think the urge means "I'm broken" or "I really do want this thing," you have to fight not just the urge but also a story about yourself. If you understand the urge as a neural reflex — like a sneeze you didn't ask for — you can step back, watch it, and run your protocol without making it about your worth as a person.

Recent extensions of this work show the same wanting-liking split in **internet gaming disorder** and **tobacco use disorder** (Ma et al., 2024), and in **gambling and food addictions** (Robinson, Fischer, Ahuja, Lesser, & Maniates, 2016). The pattern isn't just about drugs — it shows up wherever a behavior gets compulsively chased without much joy.

3. Cravings Run on Mental Pictures (And You Can Crash Them)

The theory: cravings need a mental "screen"

In 2005, three psychologists — David Kavanagh, Jackie Andrade, and Jon May — proposed an idea called **Elaborated Intrusion Theory** (Kavanagh, Andrade, & May, 2005). It says cravings happen in two stages:

1. **Stage 1 — The intrusion.** A thought pops in unbidden. "Cigarette." "Drink." "Phone." This part is fast and automatic.
2. **Stage 2 — The elaboration.** Your mind *runs with it*. You picture the cigarette, you imagine the first sip, you can almost taste the food. This is where the craving gets its power.

Here's the catch: that **vivid imagery has to live somewhere**. It lives in a part of working memory called the **visuospatial sketchpad** — basically, the "mental screen" you use to picture things. And that screen has limited space.



Mental RAM

Think of your brain's imagery system like a computer's RAM. You can only run so many programs at once. If you try to picture a cigarette in vivid detail AND solve a Rubik's cube AND keep a map of your apartment in your head, something has to give. The mental "screen" can only display so much at a time.

This leads to a testable prediction: if you give your brain a different visual-spatial task to chew on, the craving image should get fuzzier and weaker — because both can't have the screen at the same time.

The evidence — and it's strong

Researchers have run this experiment many times, with different tasks and different cravings:

- **Eye movements, visual noise, and tapping patterns** all reduced food cravings in dieters and non-dieters (Kemps, Tiggemann, Woods, & Soekov, 2004).
- **Visual tasks worked; matched-difficulty auditory tasks didn't.** Harvey, Kemps, and Tiggemann (2005) found this directly — the effect isn't just "distraction." It has to be visual-spatial distraction, because that's the specific resource cravings use.
- **Clay modeling** beat counting backwards for reducing cigarette cravings (May, Andrade, Panabokke, & Kavanagh, 2010). Shaping something with your hands hijacks the same visual-spatial system.
- **Cravings selectively weaken visuospatial working memory**, not other types of memory (Tiggemann, Kemps, & Parnell, 2010). When researchers gave people three different memory tests during cravings, only the visuospatial one suffered. This is the smoking gun — it shows cravings specifically tap that exact mental resource.

The Tetris studies

The most famous experiments here used the video game Tetris. In one lab study, just **three minutes of Tetris** significantly cut the strength, frequency, and vividness of naturally occurring cravings (Skorka-Brown, Andrade, & May, 2014).

Then they took it out of the lab. Skorka-Brown, Andrade, Whalley, and May (2015) had 31 students carry iPods for a week. Seven times a day, the device asked them to rate any current cravings — for food, drugs, sex, exercise, anything. Some of the time, they then played 3 minutes of Tetris. The result: **cravings dropped by about 14 percentage points on average**, consistently, across the entire week, across every type of craving they tracked.

The takeaway for Phase 2 of QUM

This is probably the **strongest single piece of evidence** supporting any phase of QUM. Cravings genuinely seem to need your mental "screen" to stay vivid. Force a demanding visual-spatial task onto that screen — sketching, puzzles, Tetris, even rearranging objects in your head — and the craving image starts losing resolution. May, Kavanagh, and Andrade (2015) reviewed a full decade of this research and confirmed the pattern holds up.

Practical examples of Phase 2 tasks:

- Play Tetris or a similar shape-rotation game for 3–5 minutes
- Sketch an asymmetric geometric shape from imagination
- Mold something out of clay or putty
- Solve a sudoku, jigsaw, or spatial puzzle
- Mentally walk through your house and count the doors

What *won't* work as well: listening to a podcast, talking to someone, or counting in your head. Those use different mental resources — they don't compete with the craving image.

4. Why Endless Scrolling Hooks You — And How to Snap Out

The slot machine in your pocket

Behavioral addiction to digital feeds isn't just willpower failure. Researchers have shown that social media, video feeds, and porn sites use the same reward mechanics as slot machines. The technical term is a **variable-ratio reinforcement schedule** — rewards that show up unpredictably, after an unknown number of tries.



The slot machine logic

A vending machine gives you a soda every time you put in money. Easy to stop using. A slot machine pays out unpredictably — sometimes the third pull, sometimes the fortieth. That unpredictability is what makes it impossibly compelling. Social media feeds work the same way. Most posts you scroll past are boring. But every once in a while — bam — a really good one. Your brain learns to keep pulling the lever (keep scrolling) because the next one *might* be the payoff.

The science behind this is well documented. Andreassen, Pallesen, and Griffiths (2016) studied more than 23,000 people and built one of the standard scales for measuring social media addiction. They found addictive use was associated with lower self-esteem, narcissism, and several other vulnerability factors. The pattern looks similar to chemical addictions in many ways.

Matthias Brand and colleagues developed a comprehensive framework called the **I-PACE model** (Interaction of Person-Affect-Cognition-Execution) that explains how internet-use disorders develop (Brand et al., 2019). The short version: vulnerabilities (genes, personality, mood issues) + reactive emotions and urges + weakened self-control = a habit that takes over. Over time, the behavior shifts from "I'm doing this because I want to" (goal-directed) to "I'm doing this automatically" (habit-based).

Doom-scrolling is real, and brief exposure hurts

Buchanan, Aknin, Lotun, and Sandstrom (2021) ran a clean little experiment. They had people scroll 2–4 minutes of COVID-related news (the "doom" condition), or read about COVID-related acts of kindness, or get no news at all. Just **2–4 minutes of negative scrolling** caused measurable drops in positive mood and optimism compared to no news. Kindness-scrolling didn't have that effect. The bad-news scrolling specifically did the damage.

Brain changes — but read carefully

A large study of more than 7,600 adolescents (ages 10–13) found that higher daily social media use was associated with *lower* cortical thickness across prefrontal, temporal, occipital, and parietal regions of the brain (Nagata et al., 2026). These are the same regions involved in executive control, attention, and visual processing.

Honest caveat

This is a correlation, not proof of cause. We don't know if heavy social media use *thinned* these brain regions, or if kids with thinner regions to start with were more drawn to social media. Cross-sectional studies can't answer the chicken-or-egg question. Most of the research on digital addictions has this limitation.

The active-vs-passive distinction

Here's an interesting finding. Zhai et al. (2025) used brain imaging to compare people who *actively* use short-video apps (making content, deliberately searching) versus those who *passively* scroll. The passive scrollers showed reduced "alerting" — basically, their attentional readiness was lower. The brain's default-mode network (the "mind-wandering" network) was tilted in a way that suggests a more zoned-out state.

Why this matters for breaking the loop

Passive consumption hooks you into a kind of trance. **Active generation** — creating, sketching, writing, building, deliberately doing — pulls you back into a different mental mode (executive control). This is exactly why Phase 2 of QUM forces you to *produce* something rather than just distract yourself. You're not just changing the channel; you're switching modes from "passive receiver" back to "active agent."

Treatment evidence

A 2026 systematic review and meta-analysis of 125 studies looked at therapies for problematic digital technology use (Balhara et al., 2026). Structured psychological therapies and exercise-based programs both produced strong effects on smartphone addiction. That's relevant here: both *active engagement* approaches (the kind QUM uses) work better than just trying to passively will yourself to stop.

5. Breathing, Exercise, and Calming the Alarm System

What an urge feels like physically

When a strong urge hits, your sympathetic nervous system fires up — heart rate climbs, breathing speeds up, stress hormones (cortisol, adrenaline) rise. Your body essentially treats the urge like a threat. And here's the cruel twist: when your body is in alarm mode, your **prefrontal cortex** — the rational, deciding part of your brain — gets harder to access. You become more impulsive precisely when you most need to think clearly.

Trying to read a book during a fire alarm

You wouldn't try to study for an exam while a fire alarm is blaring. Your brain literally can't focus. An intense urge is your body's version of that alarm. Until the alarm quiets down, no amount of "just think about your goals" will help. You have to physically calm the system first.

The Polyvagal Theory question (an honest detour)

You may have heard of "Polyvagal Theory," developed by Stephen Porges. It's a popular framework that explains breathing and calming techniques in terms of activating a specific vagus nerve pathway (Porges, 2007). The theory has had huge clinical impact.

A scientific controversy worth knowing about

In 2023, the journal *Biological Psychology* published critiques that cut deep. Grossman (2023) wrote that "there is broad consensus among experts that each basic physiological assumption of the polyvagal theory is untenable." Neuhuber and Berthoud (2022) found the theory's anatomy doesn't match what we know about the vagus nerve. Doody, Burghardt, and Dinets (2023) showed the theory's evolutionary claims (reptiles vs. mammals) don't hold up either.

This **does not mean breathing exercises don't work**. They clearly do — the evidence is strong. It just means we may be working through different mechanisms than the popular story says. Always a good idea to know when a popular framework is contested.

What we actually know works

Slow breathing has well-documented effects. A systematic review by Zaccaro et al. (2018) found consistent results: breathing at slow rates (fewer than 10 breaths per minute) increases heart rate variability, raises certain calming brainwave patterns, lowers anxiety, and improves emotional comfort.

Heart rate variability biofeedback works for cravings. The strongest evidence comes from Eddie, Nguyen, Zeng, Mei, and Emery (2025), who ran a high-quality clinical trial (120 adults with substance use disorder). Eight weeks of wearable HRV biofeedback significantly reduced negative emotions, cravings, and substance use days. Most strikingly, it *broke the link* between feeling a craving and then using — people felt cravings but didn't act on them as often.

Exercise reduces cravings — but not for the reason most people think. De Jesus and Prapavessis (2018) studied 110 smokers doing 10 minutes of moderate exercise. Cravings dropped significantly. But when they checked: it wasn't the cortisol reduction doing the work (that effect wasn't statistically significant). It was the *change in mood* — both positive and negative affect — that mediated the benefit. Exercise makes you feel different, and that shift carries the craving down.

Scerbo, Faulkner, Taylor, and Thomas (2010) compared passive sitting, moderate walking, and vigorous running. Both walking and running reduced cravings, with no big difference between intensities. So you don't need to sprint — a brisk walk works.

The brain–heart connection

One reason this matters: heart rate variability (basically, how flexibly your heart rate changes beat to beat) is closely linked to how well your prefrontal cortex can do its job. Thayer, Hansen, Saus-Rose,

and Johnsen (2009) found that people with higher resting HRV perform better on executive function tasks. A recent causal experiment by Lee et al. (2025) showed that increases in cardiac vagal activity were directly linked to improved performance on a cognitive control task — and to increased activation in the prefrontal cortex.

Why Phase 1 of QUM is built this way

A short burst of movement (anything that gets your heart rate up briefly) followed by slow controlled breathing is doing two things at once:

1. The movement metabolizes the adrenaline spike and shifts your mood (the actual mechanism behind exercise's effect on cravings).
2. The slow breathing activates parasympathetic pathways that improve heart rate variability and restore prefrontal access.

The net effect: by the time you start Phase 2, your rational brain is back online. You can actually *do* the visual-spatial task. Without Phase 1, you'd be trying to solve a puzzle while a fire alarm is going off.

6. Practice Makes the Urge Smaller — How Your Brain Updates

What "extinction learning" actually is

This is the long game. Every time you experience a trigger and *don't* get the reward (didn't smoke after seeing the lighter, didn't scroll after the boredom hit), your brain quietly takes a note. Over many trials, the trigger stops automatically triggering the urge. This is called **extinction learning**.

Pavlov in reverse

You know Pavlov's dogs — bell rings, food comes, dogs eventually drool just at the bell. But here's the part most people don't know: if you keep ringing the bell *without* food, the dogs eventually stop drooling. That's extinction. Your craving triggers are like that bell. If you keep encountering them without giving in, the brain slowly stops reacting.

The honest complication

Bouton's classic research showed that extinction doesn't *erase* the original learning — it adds a new learning on top (Bouton, 2004; Bouton, Maren, & McNally, 2020). The original "trigger means reward" memory is still in there, just suppressed. This is why:

- **Time gaps can cause "spontaneous recovery."** Take a long break and the old response can pop back.

- **New environments cause "renewal."** If you got clean in rehab, the old triggers may still hit you at home.
- **Stress or a slip can cause "reinstatement."** One bad day can crash you back into old patterns.

The reconsolidation hope

In 2012, a study in *Science* by Xue and colleagues caused a stir. They showed that briefly *recalling* a drug-related memory and then *extinguishing* it within a 10-minute to 1-hour window seemed to actually *update* the original memory — not just add a layer on top. Heroin users in their study showed reduced cue-induced cravings 1, 30, and 180 days later (Xue et al., 2012).

But not so fast

The picture has gotten messier since then. Some replications failed. Struik, De Vries, and Peters (2019) found the procedure actually *backfired* for nicotine — making things worse, not better. Cahill, Wood, Everitt, and Milton (2019) showed that some of the original mechanistic claims don't hold up. Memory reconsolidation in addiction is real, but it's **not the silver bullet** the early findings hinted at.

The "extinction burst"

One thing every recovering person should know: when you start consistently not giving in, the cravings often spike *harder* at first. This is called an **extinction burst**. It's not a bad sign — it's actually a sign you're disrupting the old loop. The brain is essentially yelling "are you sure?" by ramping up the request. If you ride through it, the cravings then start fading faster than they did before.

Why repeated use of QUM matters

The "Resilience Shield"

Every time you open QUM instead of giving in to a craving, two things happen:

1. **You complete an extinction trial.** Trigger present, reward absent, life continues. Your brain logs it.
2. **You build self-efficacy.** You learn — through experience, not lectures — that you *can* get through a wave without using.

Over enough repetitions, the baseline intensity of cravings often degrades. Not because the underlying sensitization (from Section 2) is gone — Robinson and Berridge (2008) note that sensitization persists for a long time. But the *expression* of that sensitization in the moment can be managed.

Important caveat: extinction is highly context-dependent. Practicing QUM in your bedroom may not fully generalize to the office or a friend's house. The research suggests you need varied practice across *multiple settings* to build durable resilience (Conklin & Tiffany, 2002; Mellentin et al., 2017).

7. Putting It All Together

Here's the QUM logic, with the science underneath each step:

Phase 1: Calm the body (≈ 1–3 minutes)

What you do: A short burst of physical effort (push-ups, jumping jacks, brisk walking — anything that elevates your heart rate briefly), then 1–2 minutes of slow controlled breathing (try a 4-second inhale, 4-second hold, 6-second exhale).

Why it works: Exercise shifts your mood through known pathways (De Jesus & Prapavessis, 2018). Slow breathing improves heart rate variability and restores access to your prefrontal cortex (Zaccaro et al., 2018; Lee et al., 2025). You can't run the rest of QUM if your alarm system is still blaring.

Phase 2: Hijack the mind (≈ 3–5 minutes)

What you do: A visual-spatial task that demands real attention. Tetris, sketching, puzzle, clay modeling, drawing a maze.

Why it works: Cravings need the brain's "visual sketchpad" to stay vivid. A demanding visual-spatial task competes for that exact resource and crashes the craving image. This is the most empirically solid mechanism in the whole protocol (Skorka-Brown et al., 2015; Kavanagh et al., 2005).

Phase 3: Ride out the wave, build resilience

What you do: Keep breathing slowly. Notice the wave passing. Don't congratulate yourself loudly or beat yourself up — just register that you got through it.

Why it works: Each successful ride-out is an extinction trial. Over time and with varied practice, your brain updates its expectations (Bouton, 2004). The trigger gradually stops producing the same intensity of response.

The three phases address the three levels of a craving: **body** (Phase 1), **mind** (Phase 2), and **learned habit** (Phase 3). That layered design matches what we know about how cravings actually operate.

8. What This Doesn't Prove (Honest Limitations)

Good science requires being clear about what we *don't* know. Here's an honest list:

- **The integrated QUM protocol itself hasn't been tested as a single package.** Each component has evidence behind it. Whether they combine well — additively, synergistically, or maybe even canceling each other out — is still an open question.
- **Polyvagal Theory's anatomy is contested.** Breathing exercises work; the popular explanation of *why* they work is currently under serious scientific scrutiny.
- **Memory reconsolidation findings have failed to replicate in some cases.** The hope that we can permanently "rewrite" addiction memories is still preliminary and drug-specific.
- **Most studies measure short-term effects.** What happens over years, with real-world chaos and missed practice days, is less well-studied.
- **Individual differences matter a lot.** Some people respond strongly to visual-spatial interference; others may need different approaches. Predictors of who responds to what are an active research area.
- **Context-dependency limits generalization.** Extinction learned in one place may not transfer to another. Practice across many settings is probably necessary for real-world durability.

That said, the underlying science for the individual mechanisms is strong enough that a brief, smartphone-deliverable protocol like QUM is a reasonable thing to try — especially when standard alternatives (white-knuckle willpower, formal therapy access, medication) are hard to access in the moment. The research base supports the components even if the integrated protocol itself awaits formal trials.

The original technical version of this review (with full neuroscience terminology and detailed mechanistic discussions) is available as a companion document. Everything in this plain-English version is grounded in the same peer-reviewed sources listed below.

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